College Board AP® Chemistry

Your notes

Atoms & Electrons

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Atomic Structure

The Atom

- The atom is the smallest building block of matter
- It consists of three subatomic particles
  - **Protons** are **positive** particles found in the nucleus
  - **Neutrons** are **neutral** particles found in the nucleus
  - **Electrons** are **negative** particles that orbit the nucleus in electron shells
- The electrons determine the chemical properties of the element
- Neutral atoms have the same amount of **protons** and **electrons**

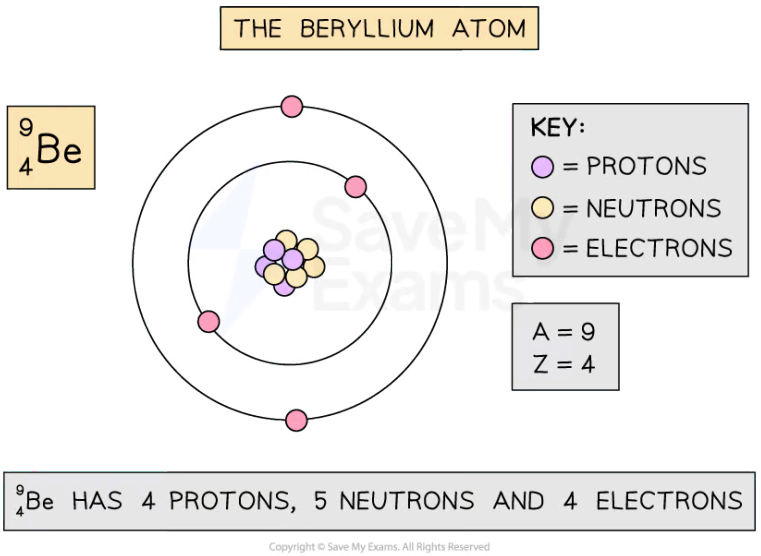
Atomic Number and Mass Number

- The **atomic number** (**Z**) of an element is the number of protons in its nucleus
  - It defines the element's identity and it always a whole number in the periodic table
    - Z = Number of protons
- The **mass number** (**A**) is the sum of protons and neutrons in the nucleus
  - It represents the atom's mass
  - Electrons **do not contribute** to the mass of the atom because they are too light compared with protons or neutrons
    - A = Number of protons + Number of neutrons
- **Isotopes** are atoms of the same element with the **same atomic number** but **different mass numbers**. Therefore, protons are the same but neutrons are different
- Elements are usually represented with the A/Z notation

$$\begin{matrix} A \\ Z \end{matrix} \text{Atomic symbol}$$

The beryllium atom

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The beryllium atom contains 4 protons, 5 neutrons and 4 electrons

Worked Example

Calculate the atomic number and mass number for an atom with 17 protons and 20 neutrons. After this, identify the element and write it down using the A/Z notation.

Answer:

- **Step 1:** Identify the given information
  - The problem provides the number of protons (17) and neutrons (20).
- **Step 2:** Determine the atomic number (Z)
  - The atomic number (Z) represents the number of protons in an atom. It also defines the element's identity.
  - Z = 17
  - So, the atomic number is 17, which corresponds to the element chlorine (Cl) on the periodic table.
- **Step 3:** Calculate the mass number (A)
  - The mass number (A) is the sum of protons and neutrons in the nucleus.
  - A = Number of protons + Number of neutrons
  - A = 17 protons + 20 neutrons
  - A = 37
  - The mass number (A) for this atom is 37.
- **Step 4:** Identify the element
  - With an atomic number of 17 (Z = 17), we can determine that the element is chlorine (Cl).
  - So, the atom in question is a chlorine atom with 17 protons, 20 neutrons, an atomic number (Z) of 17, and a mass number (A) of 37.

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- **Step 5:** Write it down with the A/Z notation

$$^{A}_{Z}\text{Atomic symbol}$$

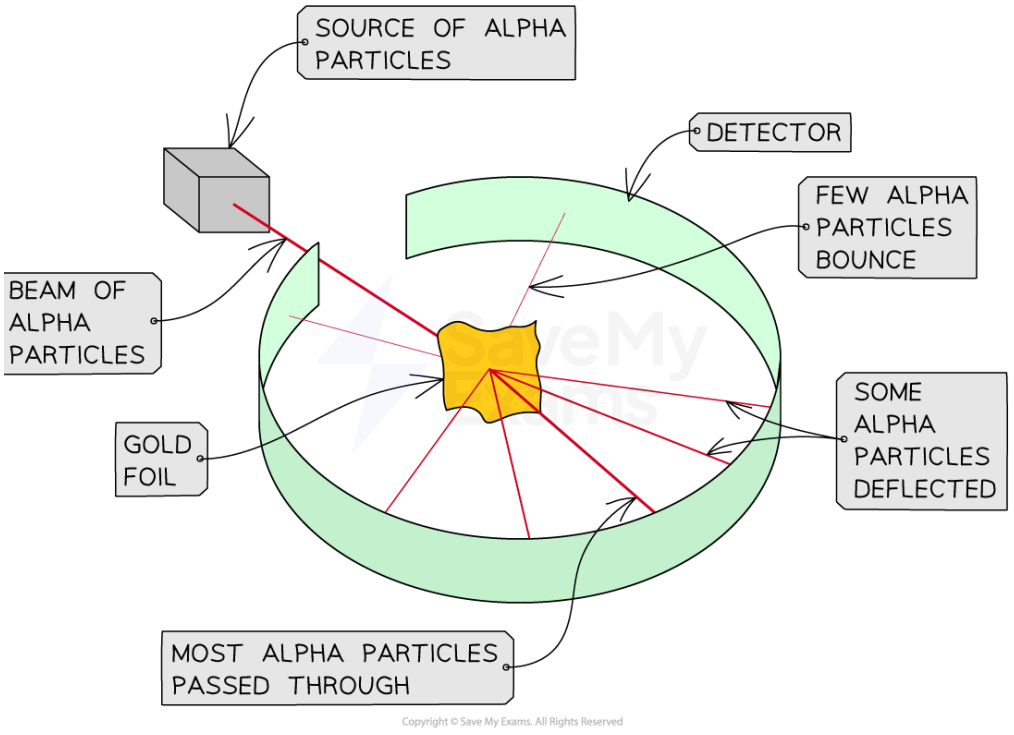
$$^{37}_{17}\text{Cl}$$

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The Rutherford Experiment

- Rutherford performed an experiment to understand the structure of the atom
- He bombarded a gold foil with tiny positive particles, called **alpha particles**
- Most alpha particles passed through the foil, but some were significantly deflected, and a few of them bounced back
- Since particles with the same charge repel, Rutherford concluded that the atoms should contain an small, dense, and positively charged core at the center
  - This core was named **nucleus**
  - The **nucleus** contain protons and neutrons

The Rutherford Experiment



The Rutherford experiment showed atoms contain a small, dense, and positively charged core at the center

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Using Coulomb's Law

- Coulomb's law describes the electrostatic force between two charged particles
- It states that the force is **directly proportional to the product of their charges** and **inversely proportional to the square of the distance between them**
  - $F_{coulombic} \propto (q_1 \times q_2)$
  - $F_{coulombic} \propto \frac{1}{r^2}$
- The formula for Coulomb's law is $F_{coulombic} = k \frac{q_1 q_2}{r^2}$, where:
  - $F_{coulombic}$ is the force between the charges in **Newtons (N)**
  - k is Coulomb's constant (approximately 8.99×10^9 N·m² C^{-2})
  - q_1 and q_2 are the value of the charges **in Coulombs (C)**, which is the unit of measurement for charge
  - r is the distance between the charges **in meters (m)**
- The force is attractive for opposite charges and repulsive for like charges

Examiner Tips and Tricks

Always verify the units before any calculation. If the units are not consistent, make sure to perform any transformation before replacing the values into the Coulomb's Law formula

Worked Example

Calculating the force that exists between a proton and an electron given the following information.

Charge of a proton = $+1.602 \times 10^{-19}$ C
Charge of an electron = -1.602×10^{-19} C
Distance between them = 1×10^{-10} m

Answer:

- **Step 1:** Write down the values and the Coulomb's law formula

$q_1 = +1.602 \times 10^{-19}$ C

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$q_2 = -1.602 \times 10^{-19}$ C

$r = 1 \times 10^{-10}$ m

$k = 8.99 \times 10^{9}$ N·m² C⁻²

$$F_{coulombic} = k \frac{q_1 q_2}{r^2}$$

- **Step 2:** Replace the value in the Coulomb's Law

$$F_{coulombic} = k \frac{q_1 q_2}{r^2}$$

$$F_{coulombic} = 8.99 \times 10^{9}\ \text{N m}^2\text{C}^{-2}\ \frac{(+1.602 \times 10^{-19}\ \text{C})(-1.602 \times 10^{-19}\ \text{C})}{(1 \times 10^{-10}\ \text{m})^2}$$

- **Step 3:** Calculate the force.

$F_{coulombic} = -2.31 \times 10^{-8}$ N

- The negative sign indicates an attractive force since proton and electron have opposite charges

Examiner Tips and Tricks

If the problem does not state the charge of the electron, you can find it in the AP Chemistry Equations and Constants that appear at the beginning of your examination. The charge of the proton has the same value but with a positive sign

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Shells & Subshells

Electrons and Light

- Electrons exist in distinct **shells** within an atom and they can interact with light
- Light can behave as a wave or as a particle
- Since light can behave as a wave, it has frequency (*v*) and wavelength (λ)
- Frequency and wavelength are related by $c = \lambda v$ where:
  - *v* is frequency
  - λ is wavelength
  - c is the speed of light which is approximately $3 \times 10^8 \text{ ms}^{-1}$
- Particles of light are called **photons**
- When a **photon is absorbed,** electrons jump **from an inner shell to an outer shell**
- When a **photon is emitted,** electrons jump **from an outer shell to an inner shell**
- The energy of the photon (*E*) is proportional to its frequency (*v*)
  - $E \propto v$
- The equation used to calculate the energy is $E = hv$ where:
  - *E* is the energy of the photon
  - *h* is the Planck's constant which is approximately $6.626 \times 10^{-34} \text{ J s}$
  - *v* is the frequency of the photon

Worked Example

Calculate the wavelength of a photon when the energy emitted after an electron transition was $4.81 \times 10^{-19} \text{ J}$

Answer:

- **Step 1:** Write down the values and the equations needed

$E = 4.81 \times 10^{-19} \text{ J}$

$h = 6.626 \times 10^{-34} \text{ J s}$

$c = 3 \times 10^8 \text{ ms}^{-1}$

$$E = hv$$

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$$c = \lambda v$$

- **Step 2:** Rearrange $E = hv$ in terms of v

$$E = hv$$

$$v = \frac{E}{h}$$

- **Step 3:** Replace the values and calculate v

$$v = \frac{E}{h}$$

$$v = \frac{4.81 \times 10^{-19}\ \text{J}}{6.626 \times 10^{-34}\ \text{J s}}$$

$$v = 7.259 \times 10^{14}\ \text{s}^{-1}$$

- **Step 4:** Rearrange $c = \lambda v$ in terms of λ

$$c = \lambda v$$

$$\lambda = \frac{c}{v}$$

- **Step 5:** Use v calculated in **Step 3** to calculate λ

$$\lambda = \frac{c}{v}$$

$$\lambda = \frac{3 \times 10^{8}\ \text{m s}^{-1}}{7.259 \times 10^{14}\ \text{s}^{-1}}$$

$$\lambda = 4.13 \times 10^{-7}\ \text{m}$$

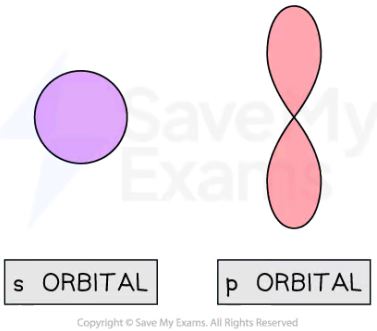
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Shells and Subshells

- Electrons are organized into **shells**
  - The farther the shell from the nucleus, the higher the energy of those electrons
- Each shell contains one or more subshells which are designated as s, p, d, and f
- Subshells are further divided into **orbitals**
- **Orbitals** are regions within a subshell where there is a **high probability** to find electrons
  - The s subshell contains one spherical orbital (spherical shape)
  - The p subshell contains three dumbbell-shaped orbitals (p_x, p_y, and p_z)
  - The d subshell contains five orbitals, and the f subshell contains seven orbitals
- Orbitals can hold a maximum of **two electrons**

s orbital and p orbital shapes

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The s orbital is spherical and the p orbital is a dumbbell shape

- The max number of electrons depends on the amount of subshells and orbitals that are held in the shell

Shell number	1	2	3	4
Subshell designation	s	s , p	s , p , d	s , p , d , f
Number of orbitals	1	1 , 3	1 , 3 , 5	1 , 3 , 5 , 7
Max number of electrons	2	8	18	32

Summary of the electronic distribution of an atom

Electron configuration

- Electron configuration is a shorthand notation that describes the distribution of electrons in an atom
- It uses the shell and subshell labels to represent the location of electrons
- Electrons in the outer shell are called **valence electrons**
  - The number of principal group in the periodic table is equal to number of valence electrons
- Electrons in the inner shell are called **core electrons**

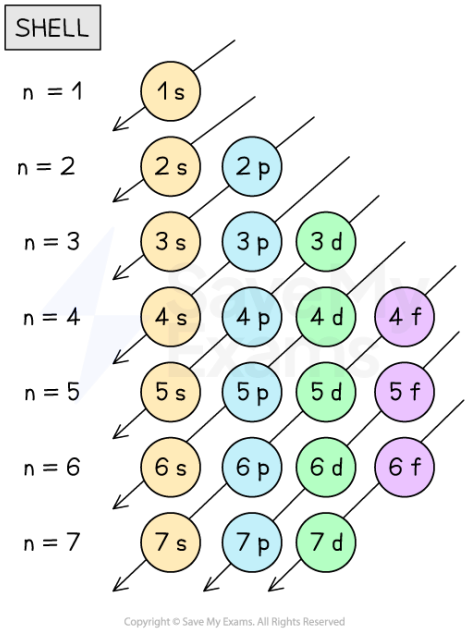
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The Aufbau Principle

- The **Aufbau Principle** helps us understand how electrons are arranged in atoms
- It states that electrons fill the lowest energy orbitals first before moving to higher energy ones
- Electrons follow the order of increasing energy
- The order of filling the subshells is determined by the arrows in **Aufbau Diagram**

The Aufbau Diagram



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Electrons fill the lowest energy orbitals first before moving to higher energy ones

Writing Electron Configurations

- Electron configuration is a way to represent the arrangement of electrons in an atom's orbitals
- Electrons occupy the lowest energy orbitals available following the Aufbau Diagram
- Each orbital can hold up to two electrons
- Electrons are represented as **exponents**
- A **condensed electron configuration** uses the closest **previous** noble gas to describe the electron arrangement



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Worked Example

Write the full electron configuration and the condensed electron configuration for fluorine (F)

Answer:

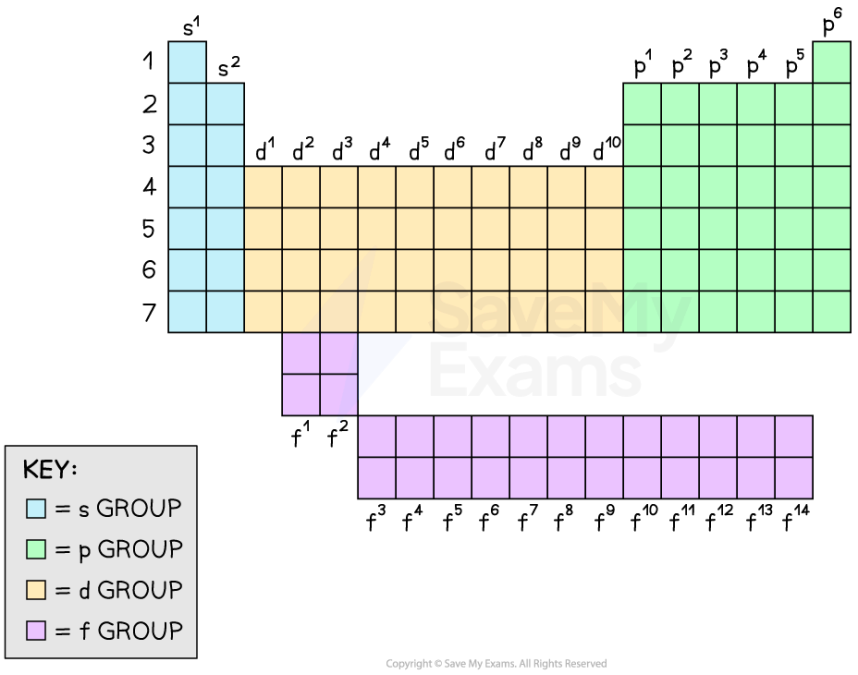
- **Step 1:** Identify the element
  - Fluorine has an atomic number of 9, which represents the number of protons. Since F is a **neutral** atom, **the number of electrons is the same as the number of protons**
- **Step 2:** Use the Aufbau Diagram to distribute the 9 electrons the electrons
  - Start by filling the orbitals with electrons following the Aufbau Diagram
  - The first two electrons will go into the 1s subshell
    $1s^2$
  - Next, move to the 2s subshell and add 2 electrons more
    $2s^2$
  - Finally, proceed to the 2p subshell. Since there are three p orbitals, the max number of electrons is 6. Considering there are just 5 electrons remaining,
    $2p^5$
- **Step 3:** Combine the individual electron configurations and interpret the information
  - $1s^2\ 2s^2\ 2p^5$
  - This tells us that fluorine has **9 electrons** arranged as follows:
    - **2 electrons** in the **1s** subshell
    - **2 electrons** in the **2s** subshell
    - **5 electrons** in the **2p** subshell
  - This also implies, that fluorine has **2 core electrons** and **7 valence electrons**
- **Step 4:** Write down the condensed electron configuration using the closest previous noble gas
  - The closest previous noble gas is helium (He)
  - The electron configuration for He is $1s^2$
  - Therefore, the condensed electron configuration for F is,
    [He] $2s^2\ 2p^5$

Electron configuration and the Periodic Table

- Elements in the **same column (group)** of the periodic table **have similar electron configurations** in their valence shell
- This similarity explains why elements in the same group exhibit similar chemical properties
- The periodic table is organized in a way that allows you to predict the electron configuration of an element based on its position in the table

The Periodic Table

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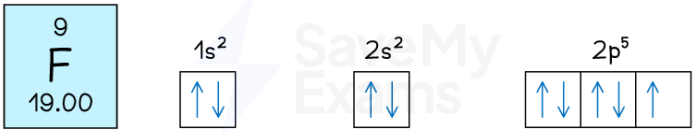
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Electron configuration in the periodic table

Orbital diagrams

- Orbital diagrams are a visual representation of electron configurations
- **Boxes** are used to represent orbitals
- **Arrows** are used to represent electrons with their spins indicated as up (↑) or down (↓)
- Each orbital can hold a maximum of two electrons with opposite spins
- Electrons will fill orbitals with the same energy one at a time before pairing up with an electron with opposite spin
- g. the orbital diagram for the atom of fluorine (F) is shown below
  - Note that an orbital is filled first with (↑) and then paired with (↓)
  - Electrons occupy first the lowest energy subshells

Representing the electron configuration in a fluorine atom



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Orbital diagram for the electron configuration of the fluorine (F) atom

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Ionization Energy

- **Ionization energy** is the energy required to remove an electron from an atom or ion in gaseous state
  - E.g. The equation for the ionization energy of sodium is shown below
- It is always positive because removing an electron always takes energy
- Coulomb's Law helps to estimate the energy based on the distance from the nucleus and the effective charge of the nucleus

Factors affecting the Ionization Energy

- **Nuclear Charge:** The greater the positive charge in the nucleus, the stronger the Coulombic attraction to electrons, leading to higher ionization energy
- **Distance from Nucleus:** Electrons closer to the nucleus experience stronger Coulombic attraction, resulting in higher ionization energy
- **Shielding (Effective Nuclear Charge):** Core electrons shield outer electrons from the full Coulombic attraction from the nucleus, reducing the ionization energy for valence electrons

Examiner Tips and Tricks

When analyzing the ionization energy between atoms, the shielding effect (effective nuclear charge) and the distance from the nucleus have more impact than the nuclear charge

Worked Example

Estimate which first ionization energy is greater, F or Cl.

Answer:

- **Step 1:** Analyze nuclear charge
  - The nuclear charge is determined by the number of protons in the nucleus
    - Chlorine (Cl): Z = 17 (17 protons)
    - Fluorine (F): Z = 9 (9 protons)
  - Chlorine has more protons (Z) than fluorine, so its nuclear charge (Z) is greater and the Coulombic attraction should be greater
- **Step 2:** Analyze distance from the nucleus

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- Chlorine has its outermost electron in the 3p subshell
- Fluorine has its outermost electron in the 2p subshell
- The distance from the nucleus is greater in Cl, therefore the electron in Cl experiences less Coulombic attraction from the nucleus

- **Step 3:** Analyzing the shielding effect (effective nuclear charge):
  - The valence electrons in Cl experience more shielding from the core electrons compared to fluorine, because there is an extra shell filled with electrons in Cl. Therefore, the effective nuclear charge is weaker in Cl
- **Step 4:** Estimate ionization energy
  - Based on the factors we've analyzed, we can make an estimation
  - Even if the nuclear charge is greater in Cl,
    - The distance from the nucleus is smaller in F
    - The effective nuclear charge is greater in F
  - Therefore, the **coulombic force of attraction** experienced by the valence electrons is **stronger in F.** This means that it requires more energy to remove a valence electron from a F atom compared to a Cl atom



Examiner Tips and Tricks

It is compulsory to make reference to the Coulomb's Law in every estimation of ionization energies between atoms



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Photoelectron Spectroscopy

- **Photoelectron spectroscopy (PES)** is an experimental technique that is used to measure the relative energies of electrons ejected by a source of radiation
- **Binding energy is** the energy required to remove an electron by overcoming the coulombic attraction to the nucleus
- A **PES spectrum** is diagram that shows the relation between the **binding energy** (x-axis) and the **relative number of electrons removed** (y-axis)
- A **PES spectrum** is directly related to the electron configuration of an atom
- **Peaks** in the PES spectrum correspond to the energy needed to remove an electron from each shell
  - Electrons in different subshells experience different **effective nuclear charge,** affecting the energy
- The **height of the peaks** in a PES spectrum determine the amount of electrons removed from a subshell

Interpreting a PES Spectrum

- Photoelectron Spectroscopy (PES) provides insight into the shells and subshells of electrons within an atom or ion
- Higher binding energy indicates electrons in subshells closer to the nucleus, which require more energy to remove
- Lower binding energy indicates electrons in subshells further from the nucleus, which require less energy to remove
- Peaks with a greater number of electrons in a subshell will be taller

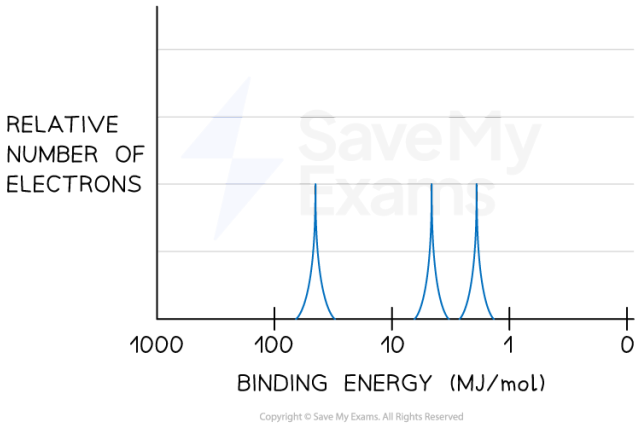
Examiner Tips and Tricks

Interpreting a PES spectrum is assessed in most of the AP Chemistry Examinations, so make sure to understand its principles and how to predict the electron configuration of an element by analyzing the spectrum

Worked Example

A PES analysis of an unknown element (**X**) was performed and the spectrum below was obtained. The sample does not contain any impurities. What is the identity of **X**?

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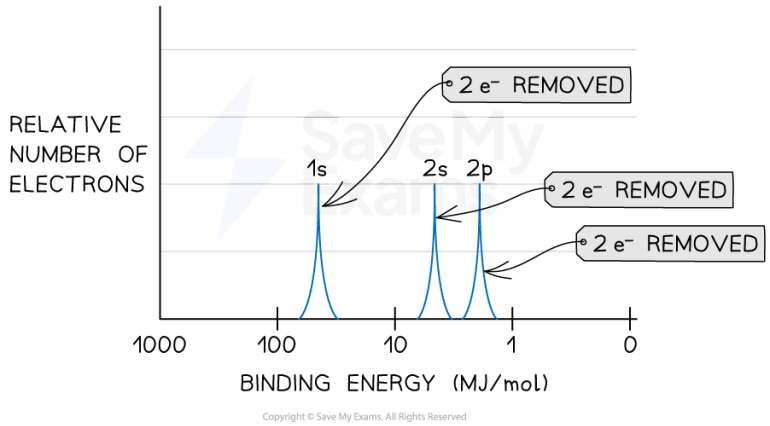


Unknown sample PES Spectrum

Answer:

- **Step 1:** Analyze the number of peaks
  - Since there are three peaks, this will correspond to the three first subshells of an atom: 1s, 2s and 2p
- **Step 2:** Analyze the position of the peaks
  - The peak with the highest binding energy will correspond to the 1s subshell since it is the closest to the nucleus, and the coulombic attractions are stronger
  - The peak with the lowest binding energy will correspond to the 2p subshell since it is the farthest to the nucleus, and the coulombic attractions are weaker
  - Therefore, the peak in the middle will correspond to the 2s subshell
- **Step 3:** Analyze the height of the peaks
  - When comparing the height of the three peaks, they are all the same.
  - Therefore, the same number of electrons have been removed in the 1s, 2s and 2p
  - Since, subshells 1s and 2s can hold up to 2 electrons, the amount of electrons in the 2p subshell for this atom was 2
- **Step 4:** Predict the electron configuration of the atom
  - $1s^2\ 2s^2\ 2p^2$
  - We can connect the electron configuration with the PES Spectrum labeled below

Your notes



PES Spectrum labeled

- **Step 5:** Identify the element using the periodic table
  - There are two ways to identify the unknown element:
  - Counting the electrons and calculating the protons
    - 6 electrons and, since it was a neutral sample, there are 6 protons

    OR
  - Counting the valence electrons and identify the period
    - There are 4 valence electrons, therefore the element should be in Group 14 of your AP Chemistry Periodic Table
    - Since the valence electrons are in the second shell, the element should be in the second period of the periodic table
- Therefore, **element X is carbon (C)**

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